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Constraining Reionization: Correlating Lyα Emission with Galaxy Properties

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Executive Summary: By the end of the funding from this grant my work will significantly transform the way we constrain the timeline of reionization. **Project 1:** Using the Texas Euclid Search for Lyman-Alpha (TESLA) survey data I have on hand **now** I will develop a predictive algorithm using thousands of galaxies between redshifts 2 – 3.5 to predict emergent Lyα emission based on SED-derived galaxy properties. **Project 2:** Performing nebular line modeling of *in-hand JWST* NIRSpec spectroscopic data of redshift 2 – 3.5 galaxies to uncover the physical properties within galaxies that regulate the escape of Lyα, providing a complete picture of how galaxies regulate Lyα emission. **Project 3:** Using an expanded TESLA data set and early *Euclid* imaging data to further constrain and robustly train my predictive algorithm from **Project 1**. **Project 4:** Directly compute the neutral hydrogen fraction using the algorithm from **Project 3** on galaxies between redshifts 6 – 10 using publicly available (and proposed-for) NIRSpec data and infer the timeline of reionization.

Introduction: Understanding the timeline of reionization can tell us about the earliest ionizing sources in the universe, and allow the study of large-scale structure formation and evolution. The timeline of reionization can be inferred by measuring the neutral hydrogen fraction during the epoch of reionization (EoR). Figure 1 shows the predictions of the neutral fraction between two competing reionization models, where reionziation is driven by bright or faint galaxies. By measuring the neutral hydrogen fraction during a period where these two models are the most different we can infer the timeline for reionization. Lyα is the most promising tracer we have at the moment as it is observed in star-forming galaxies and these galaxies are abundant in the early universe (e.g., [1]; [6]), implying that Lyα may be able to trace the spatial nature of reionization. It is a sensitive tracer, as Lyα photons are resonantly scattered, decreasing their observability when they have to traverse through neutral gas. Thus, I will use the Lyman-alpha (Lyα) emission line to measure the neutral fraction above redshift 6 to better constrain the timeline of reionization.

However, in order to use Lyα to trace reionization we need to know **how much** Lyα flux emerges from a galaxy to compare against the observed flux, as any differences are attributed to neutral hydrogen randomly scattering the Lyα photons. To date, **no good metric for acquiring the emergent Lyα flux of galaxies exists**. Getting a robust estimate of the emergent Lyα flux is complicated by processes within a galaxy, such as dust content, gas kinematics, and dust distribution which affect the Lyα strength that ultimately escapes from a galaxy (e.g., [12], [10], [3] and references therein). Previous studies [e.g. 16, 11, 8] assume that the Lyα equivalent width (EW) distribution at z $\approx$ 6 is able to predict the emergent Lyα flux of a galaxy above redshift 6. This is not a robust assumption as studies of galaxies at redshift 6 are limited by small sample sizes (typically $\sim$100-1000 sources). Spectroscopic follow-up usually requires a pre-selection of sources, which biases studies toward the most luminous sources. There are also significant differences in selection criteria between studies so every study analyzes different galaxy populations. To overcome these challenges I will create a **larger**, **unbiased-selected** sample of galaxies to tie back the emergent Lyα emission to the physical properties of an individual galaxy. I will use

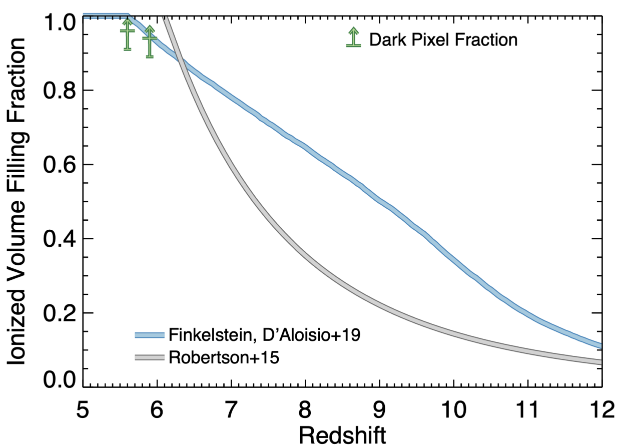

Figure 1: Figure from [5] that shows the ionized fraction as a function of redshift between competing models. The model where reionization is driven by small faint galaxies is shown in blue, while a model where bright massive galaxies drive reionization is in gray. These two models predict drastically different neutral hydrogen between redshifts 7 – 10. By measuring the neutral fraction at these redshifts via Lyα emission I will be able to infer which ionizing sources drove reionization and constrain the timeline of reionization.

the results of this analysis to train a predictive regression algorithm **to better predict a galaxy's emergent Lyα emission based on the observed properties of individual galaxies** and use this algorithm to compute the neutral fraction at $z = 6 - 10$ to **better constrain the timeline of reionization**.

Leveraging the TESLA Survey to study Lyα as a Probe of Reionization

The TESLA survey will significantly improve our ability to constrain the timeline of reionization by leveraging thousands of Lyα-emitting galaxies with a wide range of galaxy properties. TESLA is a **large**, **unbiased**, spectroscopic survey *I am leading* that is observing the 10 deg^2 *Euclid* North Ecliptic Pole (NEP) field. TESLA uses the Visible Integral-Field Replicable Unit Spectrograph (VIRUS) [7] on the Hobby Eberly Telescope. VIRUS covers 3500 – 5500 Å with 78 IFUs spanning $\sim 18' \times 18'$, allowing for optimal Lyα detection in galaxies between redshifts 1.9 – 3.5, where IGM attenuation of Lyα is minimal. The status of the TESLA survey is shown in Figure 2, where we have ∼40% of the field covered **now**, with the survey expected to be fully completed by Fall 2025 (the start of the third year of my fellowship). The NEP field has a wealth of deep multi-wavelength imaging already available from the Hawaii 20 degree (H20) survey and soon will have *Euclid* near-infrared (NIR) imaging and photometry. The H20 data includes Subaru Hyper Suprime-Cam *grizy* and *Spitzer* IRAC imaging and photometry. The H20 survey is $> 90\%$ complete, with the remaining needed nights expected to be allocated prior to the first year of funding (Spring/Summer 2023). Through collaboration with the H20 team, I have access to fully reduced imaging and photometric catalogs which are pivotal in acquiring a galaxy's global properties through SED fitting and using my emission line classification methodology [14]. This combination of H20 + TESLA allows for accurate measurements of the emergent Lyα flux from TESLA galaxies.

To properly train an algorithm to predict the emergent Lyα strength of galaxies I require a large sample of galaxies with a wide range of galaxy properties to sample a broad dynamic range in galaxy stellar population properties. While TESLA will ultimately provide 1000's of Lyα-selected galaxies, my pilot study (with 43 such galaxies) has shown that galaxies in the TESLA survey span a large dynamic range in various galaxy properties, (e.g. stellar mass, dust, SFR, and more) and that correlations do exist between galaxy properties and Lyα strength [14]. The results imply that with a larger sample of galaxies, I can leverage correlations between galaxy properties and Lyα EW to predict their emergent Lyα strength.

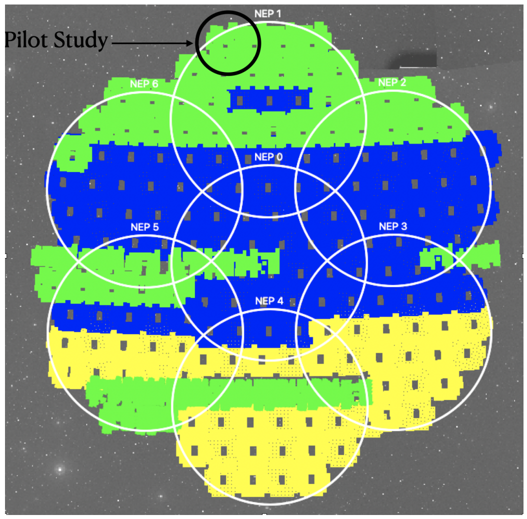

Figure 2: The 10 deg^2 North Ecliptic Pole field. My pilot study focuses on the black circle at the top of the image and consists of 2 HETDEX pointings with each pointing consisting of an 18' x 18' rectangular region being observed with ~300 IFUs (4 × 78 IFU sub-pointings). The in-hand HETDEX data is outlined by the green regions, while the white circles are the Hyper Suprime Cam-H20 imaging survey which includes deep optical imaging in the g, r, i, z, y bands. We expect that the NEP field will be > 100% covered by VIRUS (denoted by the green+blue regions) by the start of the third year of funding.

Testing Feasibility in TESLA: This past year I conducted a pilot study on a portion of the NEP field where the H20 and TESLA surveys had overlapping data, shown in the upper left of Figure 2, to test the feasibility of using correlations between galaxy properties and Lyα EW into a predictive algorithm [14]. I developed a methodology to find galaxies emitting Lyα emission and fit their stellar populations using the Bayesian SED-fitting code `BAGPIPES` to get their galaxy properties[2]. Using my methodology I discovered 43 galaxies and uncovered that the most correlated galaxy properties with Lyα EW are stellar mass and star formation rate (SFR) with a Spearman correlation coefficient of $-0.34^{+.17}_{-.14}$ and $-0.37^{+.16}_{-.14}$, indicating that these two properties can be used to predict the emergent Lyα emission of galaxies. However, a much larger sample is needed to increase the significance of the correlations, which I will do in my first year of the fellowship by significantly expanding the analysis to include more of the green VIRUS data in Figure 2, substantially increasing the sample size to >1000 Lyα-detected galaxies. The increased sample size will provide more robust correlations between galaxy properties and Lyα EW and will be used to train an algorithm to predict the emergent Lyα emission. This will result in **one published paper** that will outline the expanded sample size, methodology, and a study of their galaxy properties, and introduce the algorithm to predict emergent Lyα strength.

***Euclid**:* *Euclid*, launching in the summer of 2023, will observe the NEP field deeply over its mission. My PI is one of the few U.S. scientists to have access to proprietary *Euclid* near-infrared (NIR) YJH imaging that can be used in my TESLA analysis. The YJH filters are crucial in constraining galaxy properties as they probe the Balmer break of galaxies between redshifts 2 – 3.5, breaking the dust-age degeneracy. Prior studies have found there to be an anti-correlation between Lyα EW and dust (e.g. [19, 17, 20]), which the inclusion of *Euclid* data can improve my derived correlations between Lyα EW and dust. By the time *Euclid* completes its first year of observations in NEP, TESLA will be nearly complete – later in my fellowship I will use this increased TESLA data and *Euclid* photometry to **publish** a paper updating my predictive Lyα algorithm inclusive of the *Euclid* data and the expanded sample size. Since NASA is collaborating with *Euclid*, the NASA-funded project will play a pivotal role in better constraining the timeline of reionization by better constraining the

SED-derived properties of galaxies and more accurately calibrate the algorithm to predicting the emergent Lyα strength based on galaxy properties.

Training Algorithm Above Redshift 3.5: The analysis of TESLA will train an algorithm that is well suited to predict the emergent Lyα emission of galaxies between redshifts 2 – 3.5. However, this algorithm needs to work above redshift 6 to measure the neutral fraction during EoR. To accurately predict emergent Lyα flux from higher redshifts galaxies I will leverage existing higher redshift studies of galaxies with detectable Lyα emission to train the algorithm against (e.g. [8, 18, 4]), adding in redshift-dependent terms as needed. This is crucial in making robust measurements of the neutral fraction above redshift 6 to distinguish between competing reionization timelines. **I will use this algorithm in my 4th Project which will measure the neutral fraction between redshifts 6 – 10 (see below)**.

NEP Summary: This portion of my proposed work will result in two published papers: one outlining the correlations between galaxy properties and Lyα strength using thousands of galaxies between redshifts 2 –3.5 and introducing my algorithm to predict the emergent Lyα flux based on galaxy properties, expected early 2024. The second paper will expand upon the first paper by combining the full TESLA field including early *Euclid* YJH photometry to improve emission line identification, better constrain the SED-derived galaxy properties, and improve my predictive algorithm, expected by late 2025.

Nebular Line Modeling of 1.9 $<$ z $<$ 3.5 Galaxies using NIRSpec

Galactic spectra allow for detailed measurements of the underlying stellar population and the driving physical properties leading to observed emission and absorption lines. As a member of the Cosmic Evolution Early Release Science (CEERS) team, I have access to fully reduced *JWST* NIRSpec data **now** from CEERS Cycle 1 data. I will use NIRSpec spectra to perform nebular line modeling of galaxies in the Extended Groth Strip (EGS) field between redshifts 1.9 – 3.5 to measure the internal properties of galaxies. To study how internal galaxy properties promote Lyα escape I will use the HETDEX survey, as HETDEX is probing the Lyα emission of galaxies at these redshifts. There are $\sim$ 15 galaxies that have both HETDEX Lyα detections and CEERS NIRSpec spectra. To increase the sample, I will use any relevant guaranteed time observation data once it becomes available in 2023, as the NIRSpec team's Wide survey will also be observing several HETDEX-observed CANDELS fields. Once all the data is collected, I expect to have spectra for over 50 galaxies to perform this analysis on, and this will result in **one published paper studying the internal properties of galaxies that promote the escape of Lyα emission**.

This project will focus on exploring the internal properties of galaxies to answer ***why*** Lyα is emitting from these galaxies. The NIRSpec instrument operates between 1 – 5 microns (μm), probing the rest-frame optical of galaxies between redshifts 1.9 – 3.5. The rest-frame optical spectra of galaxies have many diagnostic emission lines that can be used to infer the physical properties of a galaxy. Figure 3 is a z $\approx$ 2 NIRSpec spectrum of a galaxy and shows prominent diagnostic emission lines that we can use to perform nebular line modeling. Instead of using empirical line ratio analysis, which has assumptions in their calculations or is correlated with other parameters [15], I will be using line ratios determined by `MAPPINGS` [9] which incorporates temperature gradients, a realistic representation of atoms, and a self-consistent theoretical model that incorporates the coupling of many parameters. I built an MCMC algorithm to robustly sample all the `MAPPINGS` parameters to determine the ISM

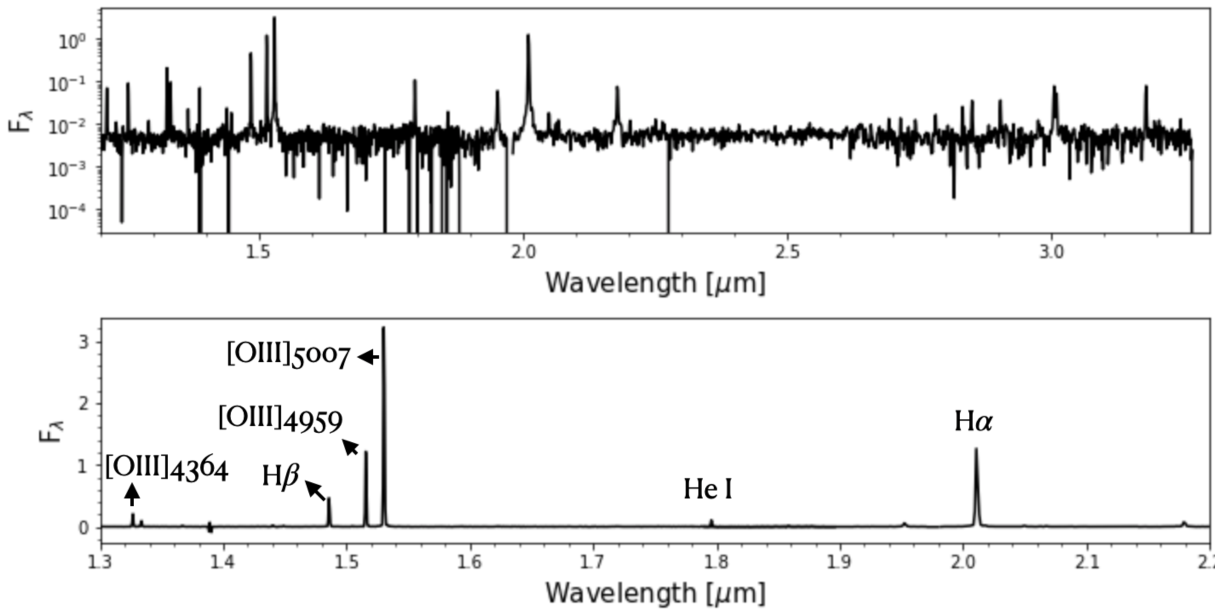

Figure 3: CEERS JWST NIRSpec spectra from a z ≈ 2.03 HETDEX LAE. We can see the wealth of lines such as Hβ, [OIII]5007, 4959, 4363, and even probing Hα emission which is useful for emission line diagnostic measurements.

properties that best reproduce the ratios of each galaxy. This is pivotal in understanding which property or set of properties aids or hinders the escape of Lyα within a galaxy. **This analysis will result in a paper explaining what properties promote or hinder the escape of Lyα, and when used with Project 1 and 3 will tell the complete story of how Lyα is able to escape from galaxies**.

Measuring the Neutral Hydrogen Fraction During Reionization

With *JWST* NIRSpec we are now able to detect Lyα emission from galaxies during EoR [18]. Lyα can be used **now** to measure a wide range of neutral fractions throughout reionization. However, the main limitation in this analysis is that no good metric for acquiring the emergent Lyα flux of galaxies exists. This will change when I complete **Project 1 and 3**, expected in late 2025, as I will have a robust algorithm to predict the emergent Lyα emission based on a galaxy's physical properties.

I will then use direct observations of $z > 6$ galaxies to apply my algorithm directly to measure neutral fractions. I will use relevant JWST NIRCam and MIRI imaging and photometry to select high redshift galaxies and use any relevant photometric catalogs to acquire their global galaxy properties through SED-fitting. To get the Lyα emission from these sources I will cross-match high redshift candidates to any relevant public NIRSpec observations, as Cycle 1 had many NIRSpec observations taken of high redshift galaxies (and Cycle 2 likely will as well). I will supplement these data sets by **leading a *JWST* Cycle 3 proposal to acquire Lyα emission**. Having a substantive high-redshift galaxy population is crucial to average out line-of-sight variations of an inhomogeneous reionization to get a sense of the average Lyα IGM transmission, and hence the neutral hydrogen fraction, at various redshift intervals. I will compare these measurements against the model predictions from Figure 1 to infer the timeline of reionization. This analysis will result in **one published paper outlining the computation of the neutral fraction as a function of redshift**

and my analysis to better contain the timeline of reionization.

We note that the PI has a current ADAP proposal to fund research in the NEP field but the FINESST proposal goes beyond what NASA has agreed to support in several ways: development of a novel technique based on SED modeling to classify detected emission lines, the inclusion of a theoretical comparison component, of early *Euclid* data to improve both classification and analysis, and the inclusion of my *JWST* spectroscopic project.

Relevance to Astrophysics Research Program

These projects are directly related to several stated objectives of NASA's science mission directive (SMD) 2014 Science Plan (e.g. see Chapter 4) as well as those listed in the 2020-2024 A Vision for Scientific Excellence strategy 1.1. The physics of the cosmos: (1) Discover how the universe works (2) Explore the origin and evolution of galaxies, stars, and planets that make up our universe, Cosmic Origins Program: (3) Understanding how the universe evolved since the Big Bang. The grant will support research that addresses the timeline of reionization which directly ties in with the evolution of the universe objective outlined above. By studying the physical properties of galaxies between z $\sim$ 1.9-3.5 we will explore any redshift-dependent properties that follow the galactic evolution objective. Once my predictive model is finished, we will be able to better understand the role galaxies played during reionization and answer how the universe evolved during this epoch. Using NASA-funded instruments such as the *JWST* to perform nebular line modeling to determine the exact physical mechanism(s) that regulate the escape of Lyα in galaxies at z $\sim$ 3 and *Euclid* NIR photometry to break the age-dust degeneracy will demonstrate the affect the NIR has on my predictive model and improve the algorithm. Both of these NASA-funded instruments will help me gather a complete picture of how Lyα escapes from Lyα-emitting galaxies.

Timeline: From the time of funding from this grant I plan on publishing **4** papers.

Year 0: Spring 2023 - Submitted the NEP Pilot study to ApJ. I will work on scaling my code to handle the current TESLA data on hand now, green pointings in Figure 2. Concurrently, I will be getting code up and running to analyze NIRSpec data of galaxies between redshifts 2 – 3.5 in existing CEERS data.

Year 1 - Start and **publish** a paper on the first NEP project using an increased sample size of $>$ 1000 galaxies and introduce the predictive algorithm. I will analyze roughly 50 HETDEX-selected galaxies that have NIRSpec spectra and work on implementing my MCMC algorithm to sample the various galaxy parameters. I will compare the galaxy properties to the Lyα emission from HETDEX to explore any correlations. At this stage, I will write the *JWST* Cycle 3 proposal to search for more galaxies with Lyα detections between redshifts 6 – 10.

Year 2 - **Publish** a paper on nebular line modeling to infer galaxy properties and correlation to Lyα emission in the NIRSpec and HETDEX data. I will then work on implementing *Euclid* NIR photometry to constrain the physical properties of galaxies and robustly train my algorithm to predict Lyα flux based on galaxy properties based on the full TESLA sample. Concurrently I will be training the algorithm using higher redshift studies to be used to measure the hydrogen fraction during reionization.

Year 3 - I will finish up any final analysis in the final NEP project and **publish** the results and predictive algorithm including *Euclid* data. I will compute the neutral hydrogen fraction during reionization and **publish** a paper on the results of this analysis, and start working on my dissertation and defense.

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Acknowledgement

The (FI) Oscar A. Chavez Ortiz authored this proposal, including the Personal Statement and the Research Statement. I received editorial support from the Principal Investigator (PI) Steven Finkelstein, my friends and colleagues Gene Leung, and Rebecca Larson, in order to improve grammar, clarity and overall structure of the proposal!!

Research Readiness Statement

Acquiring the skills necessary in order to correlate galaxy properties to Lyα emission strength has been a process I have been working on throughout my academic career. During my undergraduate studies, I made sure to take relevant courses in extra-galactic research by taking a graduate-level galaxies course to enhance my knowledge of galaxies and galaxy evolution. I also worked with Mariska Kriek on a project related to stellar synthesis modeling of galaxies in order to determine their global properties. That research experience gave me valuable scientific skills such as error propagation, spectral continuum removal, identification of emission lines, and line fitting techniques.

In the Summer of 2019 I did research to identify whether 12 low redshift galaxies ($z<0.5$) were viable candidates for being analogs of galaxies during the epoch of reionization. The task involved making my own pipeline that reduced the 2-dimensional spectral data into a 1-dimensional spectrum, continuum removal to acquire accurate flux measurements for line ratio diagnostics. I was able to calculate the equivalent width (EW) of these sources and compare them with known Lyman-continuum leakers from other low redshift studies. I found that these galaxies were great analogs of high-redshift galaxies, exhibiting properties very similar to known Lyman-continuum leakers. This work resulted in an American Astronomical Society Research Note [13]. The key takeaway from this project was using emission lines to derive galaxy properties of interest, which is going to help me with my 2nd paper using JWST NIRSpec data when performing my emission line modeling analysis.

My time during graduate school has been actively used to understand my data and utilize the resources at my disposal to help me progress in my research projects. My first year was spent learning what an LAE looked like in the TESLA dataset and learning the characteristics that separate them from [OII]-emitters. I also got familiar with the SED fitting code, `BAGPIPES`, by taking test galaxies and fitting them to acquire global galaxy properties. I learned how to utilize the resources at the Texas Advanced Computing Center (TACC) to help me achieve my research goals, such as accessing the TESLA data and running jobs on their supercomputer. At the same time, I made sure to take courses relevant to my research by taking a galaxies course, and a cosmology and large-scale structure course.

My second year in graduate school was spent automating my data process pipeline and scaling my code to handle a sample size that would be 10x larger. As of the writing of this proposal, I have in place a pipeline that is able to take a sample of VIRUS spectra, match them to an imaging counterpart, determine if the source is an LAE, and fit them with `BAGPIPES` to get galaxy properties. I also took the time to take courses designed to enhance my research project by taking an astronomical data science course, learn statistics, and Bayesian techniques. I also took radiative transport to better understand the mechanisms involved that enhance and/or hinder the escape of Lyman-alpha in a galaxy.

During my third year in graduate school, I have gotten familiar with the output from the `MAPPINGS` nebular line fitting code. I have designed an algorithm to use the results of `MAPPINGS` and run them using Markov Chain Monte Carlo (MCMC) to acquire the best set of properties for a given set of line ratios. I also worked with *JWST* NIRSpec data and used my MCMC `MAPPINGS` algorithm to find the set of parameters for every galaxy I have and see if there are any correlations that promote or hinder Lyman-alpha escape.

I believe I am capable of carrying out the research I have stated in the main proposal and on track to finish my Ph.D. in three years, by the end of the funding of this grant (08/2026).

Resume/CV

CV are mandatory and are needed for both the PI and FI, CVs are limited to 2 pages each

Current and Pending Statements

F.I. Oscar A. Chavez Ortiz

1. *Pending:* NASA FINESST
   - Asking for $50,000/year, ($40,000 salary and $10,000 for tuition expenses) for three years of funding to support me in my projects to uncover the galaxy properties that hinder and promote Lyα emission.
   - Person-months for FI: 12 months/year from 2023-2026
   - Start/End Dates: 09/01/2023 - 08/31/2026
   - Location: The University of Texas at Austin

Mentoring Plan/Agreement

Budget Timeline and Narrative

The budget justification below describes each line item in our request, which we believe is appropriate and necessary for the completion of the goals of this program over the three-year duration of this project. This budget will **begin** on **September 1st, 2023** and **end** on **August 31st, 2026**.

We request grant funding to support one full-time UT graduate student (FI O. Chavez Ortiz) for six semesters and three summers (36 months total) to carry out all of the required work on this project,as detailed in the Project Timeline section of the main proposal. This student will work at a typical appointment of 50% annual FTE effort for all three years as a graduate research assistant (GRA).

Participant Support:

1. **FI Stipend (salaries and wages):** This budget will cover one UT Austin graduate student for three full academic years spanning 12 months each (9 academic months and 3 summer months). FI's (O. Chavez Ortiz) monthly base is $3,333.33 , totaling **$40,000** for a **12-month salary**.

2. **FI Allowances (domestic travel):** No funding is requested for travel. FI (O. Chavez Ortiz) will present these scientific results at topical conferences through existing external funding sources.

3. **University Tuition:** We propose to ask that **$10,000 per year** be allocated yearly on September 1 towards tuition expenses. This allocation of funds will offset tuition costs which is a total of $19,600.00 per year (this number was based on the 2020-21 Academic calendar year). The remaining annual tuition and fees will be funded through existing external grant sources provided by the Department of Astronomy at UT (i.e. funding from PI).

4. **Other:** N/A. No funding is requested for publication costs. FI (O. Chavez Ortiz) will publish results from this work in peer-reviewed journals through existing external funding sources.

The total budget request for funds allocated to UT Austin across all three years is $150,000.

	Year 1	Year 2	Year 3
Participant Cost FI Stipend			
$40,000-12 Months	$40,000.00	$40,000.00	$40,000.00
Tuition			
$5,000 per semester, 2 semester per year	$10,000.00	$10,000.00	$10,000.00
Other			
N/A	$0.00	$0.00	$0.00
Total	$50,000.00	$50,000.00	$50,000.00